## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 1 of 13 · Question 8 part a · 3 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| 8 (a) | $2x - 75 = -31, 211$<br>$x = 22, 143$ | M1A1<br>A1<br>(3) |
| (b) | $2\frac{\sin y^\circ}{\cos y^\circ} + 5\sin y^\circ = 0$<br>$\sin y^\circ\left(\frac{2}{\cos y^\circ} + 5\right) = 0$<br>$\cos y^\circ = -\frac{2}{5}$ $(\sin y^\circ = 0)$<br>$y = 113.6^\circ$<br>$y = 0^\circ, 180^\circ$ | M1<br>M1<br><br>A1<br>B1<br>(4) |
| (c) | $3(1 - \sin^2\theta) - 3\sin^2\theta + \sin\theta + 12 = 0$<br>$6\sin^2\theta - \sin\theta - 15 = 0$<br>$(2\sin\theta + 3)(3\sin\theta - 5) = 0$<br>$\sin\theta = -\frac{3}{2}$ $\sin\theta = \frac{5}{3}$<br>As $-1 \leq \sin\theta \leq 1$ no such values for $\theta$ exist | M1<br><br>M1<br>A1<br><br>B1<br>(4) |
| | | **[11]** |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 1 of 13 · Question 8 part a · 3 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | For finding at least one correct value of $(2x-75)=-31^\circ$ or $211^\circ$<br>and attempting to find one value of $x \Rightarrow x=\frac{-31+75}{2}$ or $x=\frac{211+75}{2}$ |
| | A1 | For $x=22$ or $143$ |
| | A1 | For $x=22$ and $143$ Extra values within range – A0<br>Extra values outside of the range - ignore |
| .(b) | M1 | For using the identity $\tan y^\circ=\frac{\sin y^\circ}{\cos y^\circ}$ |
| | M1 | For factorising their expression and finding values for $\sin y^\circ$ and $\cos y^\circ$<br>$\sin y^\circ\left(\frac{2}{\cos y^\circ}+5\right)=0 \Rightarrow \sin y^\circ=0,\ \cos y^\circ=-\frac{2}{5} \Rightarrow y=...$ |
| | A1 | For $y=113.6$ if there are extra values within range – A0 |
| | B1 | For both $y=0$ and $180$ **Both required** |
| | **ALT** | |
| | M1 | For multiplying $\sin y\times\frac{\cos y}{\cos y} \Rightarrow \tan y\cos y \Rightarrow (2\tan y+5\tan y\cos y=0)$ |
| | M1 | For factorising the above expression and finding values for $\tan y^\circ$ and $\cos y^\circ$<br>$2\tan y+5\tan y\cos y=0 \Rightarrow \tan y(2+5\cos y)=0$<br>$\Rightarrow \tan y=0,\ \cos y=-\frac{2}{5} \Rightarrow y=...$ |
| | A1 | For $y=113.6$ Extra values within range – A0<br>Extra values outside of the range - ignore |
| | B1 | For both $y=0$ and $180$ **Both required** |
| | **SC** | $2\frac{\sin y}{\cos y}=-5\sin y \Rightarrow \cos y=-\frac{2}{5} \Rightarrow y=113.6$ no evidence of factorising – award M1M0A1B0 only<br>( unless there is later recovery) |
| (c) | M1 | For using the identity $\cos^2\theta+\sin^2\theta=1 \Rightarrow 3(1-\sin^2\theta)-3\sin^2\theta+\sin\theta+12=0$<br>to form a 3TQ in terms of $\sin\theta$<br>Minimally acceptable attempt is $6\sin^2\theta\pm\sin\theta\pm15=0$ |
| | M1 | For an attempt to solve their 3TQ (see general guidance for the definition of an attempt<br>$6\sin^2\theta-\sin\theta-15=0 \Rightarrow (2\sin\theta+3)(3\sin\theta-5)=0 \Rightarrow \sin\theta=...,...$ |
| | A1 | $\sin\theta=-\frac{3}{2},\ \frac{5}{3}$ |
| | B1 | For the conclusion; $\lvert\sin\theta\rvert>1$ therefore no values exist for $\sin\theta$<br>Do not accept 'undefined' without an explanation that $\lvert\sin\theta\rvert>1$ |
| **Penalise rounding only once in this question** | | |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 1 of 13 · Question 8 part a · 3 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving trig equations with a multiple or shifted angle

Answer 2 of 13 · Question 10 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

| Number | | |
|---|---|---|
| 10 (a) | $\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - \left(1 - \cos^2 A\right) = 2\cos^2 A - 1*$ | M1A1cso [2] |
| (b) | $\cos 4A = 2\cos^2 2A - 1$<br>$= 2\left(2\cos^2 A - 1\right)^2 - 1$<br>$= 2\left(4\cos^4 A - 4\cos^2 A + 1\right) - 1$<br>$= 8\cos^4 A - 8\cos^2 A + 1*$ | M1<br>M1<br>M1<br>A1cso [4] |
| (c) | Let $\frac{\theta}{4} + \frac{\pi}{24} = x$<br>$\cos^2 x\left(\cos^2 x - 1\right) = \cos^4 x - \cos^2 x$<br>$\Rightarrow \cos^4 x - \cos^2 x = \frac{\cos 4x - 1}{8}$<br>$4\left(\frac{\theta}{4} + \frac{\pi}{24}\right) = \theta + \frac{\pi}{6}$<br>$\frac{\cos\left(\theta + \frac{\pi}{6}\right) - 1}{8} = -\frac{1}{16} \Rightarrow \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$<br>$\Rightarrow \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$ | M1<br>M1<br>M1<br>M1A1 [5] |
| **Alternative** (with $x$ defined as above) | | |
| (c) | $\cos^4 x - \cos^2 x = -\frac{1}{16} \Rightarrow 8\cos^4 x - 8\cos^2 x + 1 = 1 - \frac{8}{16}$<br>$\cos 4x = \frac{1}{2} \Rightarrow 4x = \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \cdots \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$ | M1<br>M1M1<br>M1A1 |
| **Alternative** (with $x$ defined as above) | | |
| (c) | $16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0$<br>$\cos^2 x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4\times16\times1}}{2\times16} = \frac{2 \pm \sqrt{3}}{4}$<br>$\cos x = \sqrt{\frac{2 \pm \sqrt{3}}{4}}$ $(= \pm 0.9659...$ or $\pm 0.2588...)$<br>$x = 0.2617..., 1.3089..., 1.8325..., 2.8787...\left(= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}...\right)$<br>$\theta = 4\left(x - \frac{\pi}{24}\right) = \frac{\pi}{6}, \frac{3\pi}{2}$ | M1<br>M1<br>M1<br>M1A1 |
| (d) | $f(A) = \frac{\cos 4A}{2} + \frac{1}{2}$<br>$\left[\frac{\sin 4A}{8} + \frac{A}{2}\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \left(\frac{\sin 2\pi}{8} + \frac{\pi}{4}\right) - \left(\frac{\sin\frac{2\pi}{3}}{8} + \frac{\pi}{12}\right) = \frac{\pi}{4} - \frac{\sqrt{3}}{16} - \frac{\pi}{12} = \frac{\pi}{6} - \frac{\sqrt{3}}{16}$ | M1<br>dM1<br>ddM1<br>A1 [4] |
| | | **Total 15 marks** |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 2 of 13 · Question 10 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| (a) | M1 | Use $\cos(A+A)=\cos A\cos A-\sin A\sin A$ (may go straight to $\cos 2A=\cos^2 A-\sin^2 A$) and then try to replace $\sin^2 A$ by $1-\cos^2 A$<br>Allow with any other single variable. |
| | A1<br>cso | Obtain $\cos 2A=2\cos^2 A-1$ with no errors in the working.<br>Must be stated with $A$ as the variable. |
| (b) | | Allow with any other single variable for all M marks.<br>Allow one sign error removing brackets and ft for all M marks. |
| | M1 | $\cos 4A=2\cos^2 2A-1$ or $\cos 4A=\cos^2 2A-\sin^2 2A$<br>Obtain an expression in terms of cos 2A and/or sin 2A. |
| | M1 | eg $\cos 4A=2\left(2\cos^2 A-1\right)^2-1$, $\cos 4A=\left(2\cos^2 A-1\right)^2-4\left(1-\cos^2 A\right)\cos^2 A$<br>Obtain a correct expression in cos A. Eliminate terms in sin A and sin 2A. |
| | M1 | Expand brackets. Numerical factors may remain.<br>e.g. $\cos 4A=2\left(4\cos^4 A-2\cos^2 A-2\cos^2 A+1\right)-1$<br>e.g. $\cos 4A=4\cos^4 A-4\cos^2 A+1-4\cos^2 A+4\cos^4 A$ |
| | A1<br>cso | Obtain the given result with no errors in the working.<br>Must be stated with $A$ as the variable. |
| (c) | M1 | Expand brackets and use the result from part (b) to express the LHS of the equation in terms of $4x$, $\frac{\cos 4x-1}{8}$, where $x=\frac{\theta}{4}+\frac{\pi}{24}$<br>**Alternative**<br>Arrange the equation in the form $8\cos^4 x-8\cos^2 x+1=1-\frac{8}{16}$<br>Allow a single numerical or sign error in the constants for both methods. |
| | M1 | Rearrange the equation to give $\cos\left(\theta+\frac{\pi}{6}\right)=\frac{1}{2}$ or oe<br>Note that this mark is shown as B1 on ePEN. |
| | M1 | Find any correct positive value for $4\left(\frac{\theta}{4}+\frac{\pi}{24}\right)$, given in radians. eg $\frac{\pi}{3}$ |
| | M1 | Find any correct positive value for $\theta$, given in radians. eg $\frac{\pi}{6}$ |
| | A1 | Both correct answers with no extras, given in radians as multiples of $\pi$.<br>Ignore answers outside the given range. |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 2 of 13 · Question 10 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

| **Alternative** | | |
|---|---|---|
| (c) | M1 | Write the equation as a three term quadratic. e.g.<br>$16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0, \left(x = \frac{\theta}{4} + \frac{\pi}{24}\right)$,<br>$A^2 - A + \frac{1}{16} = 0, \left(A = \cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\right)$ |
| | M1 | Any valid method to solve their quadratic equation.<br>Note that this mark is shown as B1 on ePEN. |
| | M1 | Any correct value for $x = \frac{\theta}{4} + \frac{\pi}{24}$ in the range $0 \le x \le \pi$, given in radians, either as a decimal or as a multiple of $\pi$. |
| | M1 | Either value of $\theta$ given in radians,<br>$\theta = \frac{\pi}{6}$ or $\theta = \frac{3\pi}{2}$ or $\theta = 0.523...$ or $\theta = 4.712...$ |
| | A1 | Both $\theta = \frac{\pi}{6}$ and $\theta = \frac{3\pi}{2}$ given as multiples of $\pi$. |
| (d) | M1 | Use result from part (b) to write $\mathrm{f}(A) = k(\cos 4A \pm 1)$. |
| | dM1 | Integrate their $\mathrm{f}(A)$ correctly, dep on M1 awarded. Ignore limits. |
| | ddM1 | Substitute both given limits, dep on both M marks awarded. |
| | A1 | $\frac{\pi}{6} - \frac{\sqrt{3}}{16}$ *or* $\frac{1}{6}\pi - \frac{1}{16}\sqrt{3}$ simplified in this format. Do not isw. |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 2 of 13 · Question 10 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 3 of 13 · Question 7 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | |
|---|---|---|
| 7 (a) | $\dfrac{3+\left(1-\cos^2\theta\right)}{\cos\theta-2}=3\cos\theta$ | M1 |
| | $4-\cos^2\theta=3\cos^2\theta-6\cos\theta$ | M1 |
| | $(2\cos\theta+1)(2\cos\theta-4)=0$ | M1 |
| | $(\cos\theta=2$ does not exist so) $\cos\theta=-\dfrac{1}{2}$ * | A1 cso<br>(4) |
| (b) | $\cos 3x=-\dfrac{1}{2}$ | M1 |
| | $3x=120^\circ,\ 240^\circ,\ 480^\circ$ | A1 |
| | $x=40^\circ,\ 80^\circ,\ 160^\circ$ | A1 A1<br>(4) |
| | | **Total 8 marks** |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 3 of 13 · Question 7 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

| | | |
|---|---|---|
| | | Penalise any extra angles within range by deducting the final A mark. |

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 4 of 13 · Question 10 part a · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

| Number | | |
|---|---|---|
| 10 (a) | $x = \tan^{-1}(-3) = -72$ | M1 |
| | $x = 108 \qquad x = 288$ | A1 A1 |
| | | (3) |
| (b) | $7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$ | M1 |
| | $\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$ | |
| | $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$ | M1 |
| | $\tan^2\theta + \tan\theta - 6 = 0$ | A1 cso |
| | | (3) |
| (c) | $(\tan y + 3)(\tan y - 2) = 0$ | M1 |
| | $\tan y = -3 \qquad \tan y = 2$ | A1 |
| | $y = 108, 288 \qquad y = 63, 243$ | A1ft A1 |
| | | (4) |
| | | **Total 10 marks** |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 4 of 13 · Question 10 part a · 3 marks

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| Part | Mark | Guidance |
|---|---|---|
| (a) | M1 | For using inverse tan to obtain any correct angle<br>$\tan^{-1}(-3) \Rightarrow x = -71.565...^{\circ}$ Accept awrt $-72^{\circ}$ |
| | A1 | For **either** 108 **or** 288 |
| | A1 | For **both** 108 **and** 288 |
| (b) | M1 | Uses $\sin^2\theta + \cos^2\theta = 1$ on the given equation to obtain<br>$7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$ |
| | M1 | For rearranging and dividing through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain a 3TQ: $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0 \Rightarrow (\tan^2\theta + \tan\theta - 6 = 0)$ |
| | **ALT 1** | |
| | M1 | Divides the given equation through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain $7\tan^2\theta + \tan\theta = \frac{6}{\cos^2\theta}$ |
| | M1 | Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\tan^2\theta + 1 = \frac{1}{\cos^2\theta}$ and uses this result on the given equation and rearranges to achieve a 3TQ to obtain<br>$7\tan^2\theta + \tan\theta = 6(1 + \tan^2\theta) \Rightarrow (\tan^2\theta + \tan\theta - 6 = 0)$ |
| | A1 | For obtaining the given expression $\tan^2\theta + \tan\theta - 6 = 0$ * in full.<br>**Note: This is a show question, there must be no errors in the solution.** |
| | **ALT 2** | |
| | M1 | Uses the identity $\frac{\sin\theta}{\cos\theta} = \tan\theta$ with $\tan^2\theta + \tan\theta - 6 = 0$ to achieve $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$ and multiplies through by $\cos^2\theta$ to obtain<br>$\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$ |
| | M1 | Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\sin^2\theta + \sin\theta\cos\theta - 6(1 - \sin^2\theta) = 0$ and rearranges to obtain $7\sin^2\theta + \sin\theta\cos\theta = 6$ |
| | A1 cso | For obtaining the given expression $7\sin^2\theta + \sin\theta\cos\theta = 6$ in full.<br>**Note: This is a show question, there must be no errors in the solution.** |
| (c) | M1 | For changing $7\sin^2 y + \sin y\cos y = 6$ to $\tan^2 y + \tan y - 6 = 0$ [this step must be correct] and then attempting to solve the 3TQ by any method. |
| | A1 | For $\tan y = -3$ and $\tan y = 2$ |
| | A1ft | For **both** $y = 108$ **and** 288 (ft from (a)) **Do not ft angles out of range** |
| | A1 | For **both** $y = 63$ **and** 243 |
| **Rounding errors:** Penalise rounding only once in this question when first seen provided angles round to 108, 288, 63 or 243 | | |
| **Extra angles:** Deduct one A mark for any extra angles within range. Ignore angles outside of range. | | |

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| Question number | Scheme | Marks |
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Type MS · Solving trig equations with a multiple or shifted angle

Answer 5 of 13 · Question 10 part a · 3 marks

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| Number | | |
|---|---|---|
| 10 (a) | $\frac{1}{2}+\sin 3x=0 \Rightarrow \sin 3x=-\frac{1}{2} \Rightarrow 3x=-\frac{\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6} \Rightarrow x=\frac{7\pi}{18},\frac{11\pi}{18}$ | M1 |
| | Coordinates of $M$ are $\left(\frac{7\pi}{18},\ 0\right)$* | A1 |
| | Coordinates of $N$ are $\left(\frac{11\pi}{18},\ 0\right)$ | A1 [3] |
| | **ALT** | |
| | $\frac{1}{2}+\sin\left(3\times\frac{7\pi}{18}\right)=0$ | [M1 |
| | Coordinates of $M$ are $\left(\frac{7\pi}{18},0\right)$ * | A1 |
| | Coordinates of $N$ are $\left(\frac{11\pi}{18},0\right)$ | A1] |
| (b) | $\frac{dy}{dx}=3\cos 3x=0 \Rightarrow 3x=\frac{\pi}{2} \Rightarrow x=\frac{\pi}{6}$ | M1A1 |
| | $y=\frac{1}{2}+\sin 3\left(\frac{\pi}{6}\right)=1.5$ | dM1A1 [4] |
| | $\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ 1.5\right)\right]$ | |
| | **ALT** | |
| | Max of sine curve is 1 so that $\frac{1}{2}+1=\frac{3}{2} \Rightarrow y=\frac{3}{2}$ | [M1A1 |
| | $\sin 3\theta = 1 \ \Rightarrow\ 3\theta=\frac{\pi}{2} \Rightarrow\ \theta=\frac{\pi}{6}$ | dM1A1] |
| | $\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ \frac{3}{2}\right)\right]$ | |
| (c) | Uses the given $x$ coordinate for point $M$ | |
| | At point $M$ $\quad\frac{dy}{dx}=3\cos\left(3\times\frac{7\pi}{18}\right)=-\frac{3\sqrt{3}}{2}$ | B1 |
| | $y-0=-\frac{3\sqrt{3}}{2}\left(x-\frac{7\pi}{18}\right) \Rightarrow 12y+18\sqrt{3}x-7\sqrt{3}\pi=0$ o.e. | M1A1A1 [4] |
| (d) | $A=\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx+\left\lvert\left[\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx\right]\right\rvert$ | M1 |
| | $A=\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_0^{\frac{7\pi}{18}}+\left\lvert\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\right\rvert$ | M1 |
| | $A=\left[\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{\cos 3\left(\frac{7\pi}{18}\right)}{3}\right)-\left(0-\frac{\cos 3\times 0}{3}\right)\right]$ | M1 |

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| | | |
|---|---|---|
| | $+\left\|\left[\left(\frac{1}{2}\times\frac{11\pi}{18}-\frac{cos\,3\left('\frac{11\pi}{18}'\right)}{3}\right)-\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$<br>$A=[(1.23287)+(0.22828)]=1.46115\ldots\approx 1.46$ | A1<br>[4] |
| | | **Total 15 marks** |

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| | | |
|---|---|---|
| | **A1** | For correctly obtaining $x = \frac{\pi}{6}$ |
| **(c)** | **B1** | For correct gradient at point $M$<br>$\left(\frac{dy}{dx}\right) = -\frac{3\sqrt{3}}{2}$ |
| | **M1** | For a fully correct method of finding the equation of the tangent to $C$ at $M$<br>$y - 0 = ' -\frac{3\sqrt{3}}{2} ' \left(x - \frac{7\pi}{18}\right)$<br>Allow use of their $-\frac{3\sqrt{3}}{2}$ obtained from substitution of $x = \frac{7\pi}{18}$ into their $\frac{dy}{dx}$. |
| | **A1** | For a correct equation of the tangent in any form:<br>$y - 0 = -\frac{3\sqrt{3}}{2}\left(x - \frac{7\pi}{18}\right)$ |
| | **A1** | For correct equation of the tangent in the required form:<br>$12y + 18\sqrt{3}x - 7\sqrt{3}\pi = 0$<br>Accept integer multiples of this equation. |
| **(d)** | **M1** | For identifying the correct limits to find the area above the curve and the area below the curve:<br>$\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) + \left|\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)\right|$ or $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) - \int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)$<br>Allow use of their $\frac{11\pi}{18}$. Needs to correctly indicate dealing with areas above and below axis. |
| | **M1** | For an attempt to integrate $\frac{1}{2} + \sin 3x$ obtaining:<br>$\frac{1}{2}x - k\cos 3x$ where $k \neq -3$<br>For this mark ignore incorrect / absent limits. |
| | **M1** | For substituting limits correctly into their integrated expression (must be a changed expression).<br>$A = \left[\left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right) - \left(0 - \frac{cos\,3 \times 0}{3}\right)\right]$<br>$+ \left\|\left[\left(\frac{1}{2} \times \frac{11\pi}{18} - \frac{cos\,3\left(\frac{'11\pi'}{18}\right)}{3}\right) - \left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$<br>Allow use of their $\frac{11\pi}{18}$. Must show substitution or M0. |
| | **A1** | For correctly obtaining awrt 1.46 |

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| Question number | Scheme | Marks |
|---|---|---|

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### Answer 6 of 13 · Question 7 part a · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| 7 (a) | $(3x - 15 =)$ 30 **or** 330 **or** 390 | M1 |
| | $x = 15$ $x = 115$ $x = 135$ | M1 A1 A1 [4] |
| (b) | $3\frac{\sin y}{\cos y} + 4\sin y = 0 \rightarrow \sin y\left(\frac{3}{\cos y} + 4\right) = 0$ | M1 |
| | $\sin y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$ | A1ft |
| | $y = -180, 0$ | A1 |
| | $y = 138.6, -138.6$ | A1 |
| | ALT | |
| | $3\tan y + 4\tan y \cos y = 0 \rightarrow \tan y\,(3 + 4\cos y) = 0$ | M1 |
| | $\tan y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$ | A1ft |
| | $y = -180, 0$ | A1 |
| | $y = 138.6, -138.6$ | A1 [4] |
| (c) | $\cos\theta = 3\,(1 - \cos^2\theta) - 1$ $\rightarrow$ $3\cos^2\theta + \cos\theta - 2 = 0$ | M1 |
| | $(3\cos\theta - 2)(\cos\theta + 1)$ | M1 |
| | $\theta = -180$ | A1 |
| | $\theta = 48.2, -48.2$ | A1 [4] |
| | | **Total 12 marks** |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 6 of 13 · Question 7 part a · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | For any of 30 **or** 330 **or** 390<br>May be implied by correct answers. |
| | M1<br><br><br>A1 A1 | Solves a linear equation coming from attempt at use of inverse trigonometric function to obtain one value.<br>e.g. solves $3x - 15 = "30"$<br>First A1 for any correct value, second A1 for all 3 correct values and no others in the range.<br>Ignore values outside the range. |
| (b) | M1 | Correctly replaces the identity for tan $y$ and attempts to deal with $\sin y$.<br>Allow for factorising. Condone dividing through by $\sin y$.<br>Minimally acceptable attempt for factorisation is $A \sin y\left(\frac{B}{\cos y}+C\right)$ |
| | A1ft | $\sin y = 0$ and $\cos y = -\frac{B}{C}$, follow through their $B$ and $C$ **only.** |
| | A1 | From sine: Both values, ignore extra values out of range, A0 for extra values in range. |
| | A1 | From cosine: Both values, ignore extra values out of range, A0 for extra values in range.<br>$y$ = awrt 138.6, awrt – 138.6 |
| | ALT | |
| | M1 | Correctly replaces the identity for tan $y$ and attempts to deal with $\tan y$.<br>Allow for factorising. Condone dividing through by $\tan y$.<br>Minimally acceptable attempt for factorisation is $A\tan y\,(B + C\cos y)$ |
| | A1ft | $\tan y = 0$ and $\cos y = -\frac{B}{C}$, follow through their $B$ and $C$ **only.** |
| | A1 | From tangent: Both values, ignore extra values out of range, A0 for extra values in range. |
| | A1 | From cosine: Both values, ignore extra values out of range, A0 for extra values in range.<br>$y$ = awrt 138.6, awrt – 138.6 |
| (c) | M1 | Correctly uses the identity for $\cos^2\theta$ and rearranges to get a 3TQ<br><br>Minimally acceptable attempt is $\pm 3\cos^2\theta \pm \cos\theta \pm 2 = 0$ |
| | M1 | Solves a 3TQ to arrive at 2 distinct values for $\cos\theta$. See general guidance. |
| | A1 | For – 180, ignore extra values out of range, A0 for extra values in range. |
| | A1 | Both values, ignore extra values out of range, A0 for extra values in range.<br>$\theta$ = awrt 48.2, awrt – 48.2 |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 6 of 13 · Question 7 part a · 4 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 7 of 13 · Question 1 part c · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 1 (a) | $t = \frac{10}{3}$<br>$P = 3 + 2\sin\frac{5\pi}{4}$<br>$= 3 - \sqrt{2}$ oe (e.g. $3 - \frac{2}{\sqrt{2}}$) | M1<br>A1<br>(2) |
| (b) (i) | 5 | B1 |
| (ii) | 1 | B1<br>(2) |
| (c) | $4 = 3 + 2\sin\left(\frac{3\pi t}{8}\right)$<br>$\frac{1}{2} = \sin\left(\frac{3\pi t}{8}\right)$<br>$\frac{\pi}{6} = \left(\frac{3\pi t}{8}\right)$<br>$t = \frac{4}{9}$ oe | M1<br>M1<br>A1<br>(3) |
| | | [7] |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Correct substitution of $t = \frac{10}{3}$, leading to a value for P, simplification not required. |
| | A1 | Answer stated, oe exact value. |
| (c) | M1 | Correctly substitutes the value of $P = 4$ and rearranges to give an expression of the form $a = \sin\left(\frac{3\pi t}{8}\right)$<br>**Do not allow this mark if $a > 1$ or $a < -1$** |
| | M1 | Correctly uses the inverse sin function to arrive at $b = \left(\frac{3\pi t}{8}\right)$ and solves to find a value of $\frac{3\pi t}{8}$, allow this value to be in degrees<br>If the inverse sin function is not shown, then the value of the angle obtained must be correct for their $b$. |
| | A1 | oe |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 7 of 13 · Question 1 part c · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 8 of 13 · Question 10 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

| Question number | | |
|---|---|---|
| 10 (a) | $(x+\frac{\pi}{3})=\frac{\pi}{3}$ or $\frac{2\pi}{3}$ or $\frac{7\pi}{3}$<br>$(x+\frac{\pi}{3})=\frac{\pi}{3}$ and $\frac{2\pi}{3}$ and $\frac{7\pi}{3}$<br>$x=0,\frac{\pi}{3},2\pi$ | M1<br>A1<br>A1<br>(3) |
| (b) | $\tan\theta=-\frac{5}{3}$<br>$\theta = -59^\circ, -239^\circ, 121^\circ, 301^\circ$ | M1<br>M1 A1<br>(3) |
| (c) | $1+\sin 2y-2(1-\sin^2 2y)=0$<br>$2\sin^2 2y+\sin 2y-1=0$<br>$(\sin 2y+1)(2\sin 2y-1)=0$<br>$\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$<br>$2y=-90^\circ, (30^\circ), (150^\circ), -330^\circ, -210^\circ$<br>$y=-45^\circ, -105^\circ, -165^\circ$ | M1<br>A1<br>dM1<br>A1<br>A1<br>(5)<br>**[11]** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Any one of the three indicated angles, radians only, ignore any other angles. |
| | A1 | For all three indicated angles, ignore other angles out of the range $\frac{\pi}{3}\le x+\frac{\pi}{3}\le\frac{7\pi}{3}$ |
| | A1 | For all three angles, ignore angles out of range, A0 if additional angles in range. |
| (b) | M1 | For $\tan\theta=k$. $k\neq0$, $k\neq\pm1$ |
| | M1 | Any one correct value, does not need to be to the nearest degree. Allow one correct value to imply the first M1. Ignore any other angles. |
| | A1 | For all four angles, ignore angles out of range, A0 if additional angles in range. All four angles must be given to the nearest degree. |
| (c) | M1 | For the correct use of $1-\sin^2 2y$ in the equation on the left or right side, equation doesn't need to be = 0. |
| | A1 | Correct 3TQ, must be = 0 and a valid attempt to solve leading to sin $2y$ = |
| | dM1 | $\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$ (allow sin2y = $a$ and $b$ from a valid attempt to solve their 3TQ). Allow sin2$y$ to be $x$ or any other variable. |
| | A1 | For minimum of 3 of the 5 values shown, (including the ones in brackets), ignore other angles outside the range -360° ≤ $2y$ ≤ 360°. Allow sight of 270 if -90° present. |
| | A1 | For all 3 values shown. Ignore extras out of range.<br>**Rounding answers (where accuracy is specified in the question)**<br>Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 8 of 13 · Question 10 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving trig equations with a multiple or shifted angle

Answer 9 of 13 · Question 7 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| **7(a)** | $\left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}} = 0 \Rightarrow \cos 3\theta = -\sqrt{3}\sin 3\theta \Rightarrow \tan 3\theta = -\frac{1}{\sqrt{3}}$ | M1 |
| | $\Rightarrow 3\theta = -\frac{\pi}{6}$ **or** $\frac{5\pi}{6}$ | A1 |
| | $\Rightarrow m = -\frac{\pi}{18}$ $n = \frac{5\pi}{18}$ | A1 [3] |
| **(b)** | $V = \pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$ | M1 |
| | $V = \pi\left[\frac{\sin 3\theta}{3} - \frac{\sqrt{3}\cos 3\theta}{3}\right]_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}$ | M1 |
| | $V = \pi\left[\left(\frac{\sin 3\left(\frac{5\pi}{18}\right)}{3} - \frac{\sqrt{3}\cos 3\left(\frac{5\pi}{18}\right)}{3}\right) - \left(\frac{\sin 3\left(-\frac{\pi}{18}\right)}{3} - \frac{\sqrt{3}\cos 3\left(-\frac{\pi}{18}\right)}{3}\right)\right]$ | M1 |
| | $V = \frac{4\pi}{3}$ | A1 [4] |
| | | **Total 7 marks** |

| Part | Mark | Notes |
|---|---|---|
| (a) | M1 | Sets the equation = 0 and obtains $\tan 3\theta = k$ |
| | A1 | Finds at least one correct value of $3\theta$<br>$3\theta = -\frac{\pi}{6}$ **or** $\frac{5\pi}{6}$<br>Works in degrees<br>Accept $-30^\circ$ **or** $150^\circ$ for this mark.<br>This mark is also implied by one correct solution for $m$ or $n$ in degrees or radians.<br>**NB This is an M mark in Epen** |
| | A1 | For $m = -\frac{\pi}{18}$ $n = \frac{5\pi}{18}$ which must be in radians.<br>Accept embedded in coordinates.<br>$M$ and $n$ do not need to be identified. |

Type MS · Solving trig equations with a multiple or shifted angle

Answer 9 of 13 · Question 7 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| (b) | **Working in degrees**.<br>Allow working in degrees up to the last M mark. | |
| | M1 | For a correct statement for the volume of revolution with $\pi$ and their limits.<br>Allow<br>$V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\left[\cos 3\theta+\sqrt{3}\sin 3\theta\right]^{\frac{1}{2}}\right)^2 \mathrm{d}\theta$ or $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$<br>Allow also:<br>$V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}x$ or even $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)$<br>**This mark can be implied by correct further working**<br>Working in degrees<br>$V=\pi\int_{-10^\circ}^{50^\circ}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$ |
| | M1 | For an acceptable attempt at integration.<br>Minimally acceptable integration is as follows.<br>$\cos 3\theta \Rightarrow \pm\frac{\sin 3\theta}{3}$, $\sin 3\theta \Rightarrow \pm\frac{\cos 3\theta}{3}$<br>Ignore absence or incorrect limits and the absence of $\pi$ for this mark. |
| | M1 | For substitution of the correct limits into their integrated expression the correct way around. This must be a changed expression from the one given.<br>If the integrated expression is correct with correct limits, allow a final volume of $\frac{4\pi}{3}$ seen without explicit substitution. If the final volume is incorrect without evidence of explicit substitution award M0.<br><br>If the integrated expression is incorrect or the limits are incorrect, explicit substitution **must** be seen for the award of this mark.<br><br>**Do not** allow use of degrees at this stage.<br><br>Ignore absence of $\pi$ for this mark. |
| | A1 | For the correct volume $(V)=\frac{4\pi}{3}$ |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 9 of 13 · Question 7 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 10 of 13 · Question 7 part b · 3 marks

2022 JUN · Paper 02 · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Solving trig equations with a multiple or shifted angle

Answer 11 of 13 · Question 11 part c · 4 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **11(a)** | $\cos 2A = \cos^2 A - \sin^2 A \Rightarrow \cos 2A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$<br>$2\sin^2 A = 1 - \cos 2A \Rightarrow \sin^2 A = \frac{1}{2}(1-\cos 2A)$ * | M1M1<br>A1cso<br>[3] |
| **(b)** | $\cos^2 A = \frac{1}{2}(1+\cos 2A)$<br>$\sin^4 x + \cos^4 x = \left(\left(\frac{1-\cos 2x}{2}\right)^2 + \left(\frac{1+\cos 2x}{2}\right)^2\right)$<br>$= \frac{1}{4}\left((1-\cos 2x)^2 + (1+\cos 2x)^2\right)$<br>$= \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x + 1 + 2\cos 2x + \cos^2 2x\right)$<br>$= \frac{1}{4}\left(2 + 2\cos^2 2x\right) = \left\{\frac{1}{2}\left(1+\cos^2 2x\right)\right\}$<br>$\left\{\cos^2 2x = \frac{1+\cos 4x}{2}\right\}$<br>$\sin^4 x + \cos^4 x = \frac{1}{2}\left(1 + \frac{1+\cos 4x}{2}\right) = \frac{3+\cos 4x}{4}$ * | B1<br><br><br>M1<br>A1<br><br><br>M1A1<br>cso<br>[5] |
| **(c)** | $5\sin 2\theta + 6 = 8\left(\frac{3+\cos 2\theta}{4}\right) \Rightarrow 5\sin 2\theta + 6 = 6 + 2\cos 2\theta$<br>$\Rightarrow 5\sin 2\theta = 2\cos 2\theta \Rightarrow \frac{\sin 2\theta}{\cos 2\theta} = \frac{2}{5} \Rightarrow \tan 2\theta = \frac{2}{5}$<br>$2\theta = 21.801...^\circ, 201.801...^\circ, 381.801...^\circ$<br>$\Rightarrow \theta = 10.9^\circ, 100.9^\circ$<br>Penalise extra angles in range by withholding the final A mark.<br>Extra angles out of range – ignore. | M1<br>M1<br><br>A1<br>A1<br>[4] |
| | | **Total 12 marks** |

| Part | Mark | Notes |
|---|---|---|
| (a) | M1 | For using the summation formula for cos 2A<br>They must start with either<br>$\cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A \cos A - \sin A \sin A$ |
| | M1 | For eliminating $\cos^2 A$ using $\cos^2 A + \sin^2 A = 1$ and attempting to rearrange to the required result. This is not dependent on the first M mark, so if they start with another identity for cos2A they can still get this mark. |
| | A1<br>cso | For the correct identity with no errors seen.<br>This is a given result. |
| | **NB** Some candidates work backwards – that is fine, please follow their working. | |

Type MS · Solving trig equations with a multiple or shifted angle

Answer 11 of 13 · Question 11 part c · 4 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **Working with a different variable.**<br>If they work in this part with a different variable (eg $A$) then award all the marks as appropriate up to the last mark.<br>If they leave their final answer in terms of another variable, withhold the final A mark only.<br>If they however, change to $x$ on the final line award all the marks [provided everything is correct] | | |
| (b) | **Main method** | |
| | B1 | For use of the correct identity for $\cos^2 x$<br>$\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$ or if they convert $\cos^2 x$ to $\sin^2 x$ use the Pythagorean identity $\cos^2 x = 1-\sin^2 x$ and apply the given identity. |
| | M1 | For squaring both identities. This must be a correct expansion.<br>Ft their identity for $\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$ |
| | A1 | For collecting like terms and obtaining $\frac{1}{2}(1+\cos^2 2x)$ oe.<br>For example, $\frac{1}{4}(2+2\cos^2 2x)$ |
| | M1 | For applying the identity:<br>$\cos^2 A = \frac{1}{2}(1+\cos 2A) \Rightarrow \left[\cos^2 2A = \frac{1}{2}(1+\cos 4A)\right]$<br>on $\cos^2 2x$ **only** again to achieve an expression in $\cos 4x$ **only** |
| | A1<br>cso | For the correct identity with no errors seen.<br>This is a given result. You must check every line of their work carefully. |
| | **ALT 1** | |
| | B1 | For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working]. |
| | M1 | For using the expansion of $\sin^2 x+\cos^2 x$ as follows<br>$\sin^4 x+\cos^4 x = (\sin^2 x+\cos^2 x)^2 - 2\sin^2 x\cos^2 x$<br>This must be correct |
| | A1 | For obtaining $\sin^4 x+\cos^4 x = 1-\frac{1}{2}\sin^2 2x$ oe.<br>For example; $\frac{1}{2}(2-\sin^2 2x)$ |
| | M1 | For applying the given identity on $\sin^2 2x$ **only** to achieve an expression in $\cos 4x$ **only**<br>$\sin^2 A = \frac{1}{2}(1-\cos 2A) \Rightarrow \left[\sin^2 2A = \frac{1}{2}(1-\cos 4A)\right]$<br>$\sin^4 x+\cos^4 x = 1-\frac{1}{2}\left(\frac{1-\cos 4x}{2}\right) \Rightarrow \left(\sin^4 x+\cos^4 x = \frac{3+\cos 4x}{4}\right)$ |
| | A1<br>cso | For the correct identity with no errors seen.<br>This is a given result. You must check every line of their work carefully. |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 11 of 13 · Question 11 part c · 4 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| | **ALT 2** – Works backwards from the given result | |
| | B1 | For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working]. |
| | M1 | Applies the cos2A identity and converts 3 into $3\left(\sin^2 2x+\cos^2 2x\right)$<br>$\frac{3+\cos 4x}{4} = \frac{3\sin^2 2x+3\cos^2 2x+\left(\cos^2 2x-\sin^2 2x\right)}{4}$ |
| | A1 | Obtains $\frac{3+\cos 4x}{4} = \frac{4\cos^2 2x+2\sin^2 2x}{4}$ |
| | M1 | Applies the cos2A identity and expands the bracket.<br>$\frac{3+\cos 4x}{4} = \frac{4\left(\cos^2 x-\sin^2 x\right)^2+8\sin^2 x\cos^2 x}{4}$<br>$= \frac{4\cos^4 x-8\sin^2 x\cos^2 x+4\sin^4 x+8\sin^2 x\cos^2 x}{4}$ |
| | A1 cso | Simplifies to the required result with no errors seen<br>$\frac{3+\cos 4x}{4} = \sin^4 x+\cos^4 x$ * |

| | | |
|---|---|---|
| (c) | M1 | For obtaining the correct equation in terms of $\sin 2\theta$ and $\cos 2\theta$<br>$6+2\cos 2\theta = 5\sin 2\theta+6$<br>Accept unsimplified, accept even: $8\left(\frac{3+\cos 2\theta}{4}\right) = 5\sin 2\theta+6$ |
| | M1 | For using the $\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta}$ identity correctly on their expression following an expression in the form $A\cos 2\theta = B\sin 2\theta$ and must be in terms of $2\theta$ |
| | A1 | For achieving at least one correct angle for $2\theta$<br>**NB This is an M mark in Epen** |
| | A1 | For awrt both 10.9° and 100.9°<br>Penalise extra angles in range by withholding the final A mark.<br>Extra angles out of range – ignore. |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 11 of 13 · Question 11 part c · 4 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

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Type MS · Solving trig equations with a multiple or shifted angle

Answer 12 of 13 · Question 10 part c · 6 marks

2023 NOV · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|
| 10(a)(i) (ii) | $\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A$ * | B1 cso |
| | $\cos(A+A)=\cos A\cos A-\sin A\sin A$ | |
| | $\Rightarrow \cos 2A=\cos^2 A-\sin^2 A$ | M1 |
| | $=\cos^2 A-(1-\cos^2 A)$ | A1*cso |
| | $=2\cos^2 A-1$ * | [3] |
| (b) | $\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$ | M1 |
| | $=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos^2\theta}-\frac{\cos^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}\times\frac{\cos^2\theta}{\cos^2\theta}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ | dM1 |
| | $=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)$ or $\left(\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}\right)=2\sin\theta\cos\theta$ | ddM1 |
| | $=\sin 2\theta$ * | A1*cso [4] |
| ALT | $\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$ | M1 |
| | $=\frac{2\frac{\sin\theta}{\cos\theta}}{1+\tan^2\theta}$ | dM1 |
| | $\cos^2\theta+\sin^2\theta=1\Rightarrow\tan^2\theta+1=\frac{1}{\cos^2\theta}\Rightarrow\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)=2\sin\theta\cos\theta$ | ddM1 |
| | $=\sin 2\theta$ * | A1*cso [4] |
| (c) | $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)}=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)\Rightarrow\frac{5}{2}\sin\left(2\left(x+\frac{\pi}{6}\right)\right)=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ | M1 |
| | $1-2\cos^2\left(x+\frac{\pi}{6}\right)=-\cos\left(2x+\frac{\pi}{3}\right)$ | M1 |
| | $\left(\frac{5}{2}\sin\left(2x+\frac{\pi}{3}\right)=-\cos\left(2x+\frac{\pi}{3}\right)\Rightarrow\right)\tan\left(2x+\frac{\pi}{3}\right)=-\frac{2}{5}$ | M1 |
| | $\left(\Rightarrow 2x+\frac{\pi}{3}=\right)-0.38050,\quad 2.76108$ | M1 |
| | $(\Rightarrow x=)-0.714,\quad 0.857$ | A1A1 [6] |

Type MS · Solving trig equations with a multiple or shifted angle

Answer 12 of 13 · Question 10 part c · 6 marks

2023 NOV · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **(d)** | $I = \int_0^{\frac{\pi}{2}} (2\sin 2\theta - \cos 5\theta + 2)\ \{\mathrm{d}\theta\}$ | M1 |
| | $= \left[\frac{-2\cos 2\theta}{2} - \frac{\sin 5\theta}{5} + 2\theta\right]_0^{\frac{\pi}{2}}$ | M1 |
| | $= \left(-\cos 2\left(\frac{\pi}{2}\right) - \frac{\sin 5\left(\frac{\pi}{2}\right)}{5} + 2\times\left(\frac{\pi}{2}\right)\right) - \left(-\cos 2(0) - \frac{\sin 5(0)}{5} + 2\times 0\right)$ | M1 |
| | $= \left(1 - \frac{1}{5} + \pi\right) - (-1 - 0 + 0) = \frac{9}{5} + \pi$ | A1 [4] |
| | | **Total 17 marks** |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 12 of 13 · Question 10 part c · 6 marks

2023 NOV · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

| Part | Mark | Notes |
|---|---|---|
| (a) | B1 cso | For $\sin(A+A)$ or $\sin 2A = \sin A\cos A + \cos A\sin A = 2\sin A\cos A$ * <br> No steps missing or incorrect work. Condone $\sin A + A$ with brackets missing if all other work correct. |
| | M1 | For $\cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A$ |
| | A1* cso | For $\cos(A+A)$ or $\cos 2A = \cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A)$ <br> $= 2\cos^2 A - 1$ <br> No steps missing, no incorrect work. Condone $\cos A + A$ with brackets missing if all other work correct. |
| Students who miss 'A' in their working, (eg coscos – sinsin) may be awarded M mark, but not A or B marks. | | |
| (b) | M1 | Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. <br> Allow the 2 to be missing for this mark **but not the final A mark.** |
| | dM1 | For $1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$ or multiplying by $\frac{\cos^2\theta}{\cos^2\theta}$ to give $\left(\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \right) \frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ <br> The $\times\frac{\cos^2\theta}{\cos^2\theta}$ does not need to be explicitly written. <br> Dependent on previous method mark. |
| | ddM1 | For minimum steps (and dependent on previous method marks): <br> $\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta} = 2\sin\theta\cos\theta$ |
| | A1*cso | Achieves the required result – minimum steps as shown, no errors. <br> **Steps in brackets do not need to be shown.** |
| ALT | M1 | Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. |
| | dM1 | For replacing $1 + \frac{\sin^2\theta}{\cos^2\theta}$ with $1 + \tan^2\theta$ <br> Dependent on previous method mark. |
| | ddM1 | For replacing $\tan^2\theta + 1 = \frac{1}{\cos^2\theta} \Rightarrow \frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}} = 2\sin\theta\cos\theta$ <br> Dependent on both previous method marks. |
| | A1*cso | Achieves the required result – minimum steps as shown, no errors. <br> **Steps in brackets do not need to be shown.** |
| Students who miss $\theta$ in their working, may be awarded M marks, but not the A mark. | | |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 12 of 13 · Question 10 part c · 6 marks

2023 NOV · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

| | | |
|---|---|---|
| Allow work throughout where candidates have let, for example $\theta = x + \frac{\pi}{6}$ and worked in $\theta$ and $2\theta$ throughout in (c) | | |
| (c) | M1 | Rearranges the given equation into $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)} = \left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ **and** changes the left hand side into $P\sin\left(2\left(x+\frac{\pi}{6}\right)\right)$ where $P$ is any constant. |
| | M1 | Correctly states or uses $1-2\cos^2\left(x+\frac{\pi}{6}\right) = -\cos\left(2x+\frac{\pi}{3}\right)$ (condone being replaced with $-\cos 2A$) |
| | M1 | For $\tan\left(2x+\frac{\pi}{3}\right) = Q$ where $Q$ is any constant. |
| | M1 | For $-0.38050$ or $2.76108$<br>This may be awarded for either value being seen, ignore any extra values seen.<br>Accept any values with at least 3 significant figures. |
| | A1 | For $(x=)-0.714$ or $(x=)0.857$<br>This may be awarded for either value being seen, ignore any extra values seen.<br>Accept any value with more than 3 significant rounding to those given. |
| | A1 | For $(x=)-0.714$ and $(x=)0.857$<br>Ignore any extra values **out of range**. Extra values in range will mean A0.<br>Accept any value with more than 3 significant rounding to those given. |
| Students must use $\theta$ or $x+\frac{\pi}{6}$ and $2\theta$ or $2\left(x+\frac{\pi}{6}\right)$ | | |
| (d) | M1 | For changing the integral into $(I=)\int_0^{\frac{\pi}{2}}(2\sin 2\theta - \cos 5\theta + 2)\,\{\mathrm{d}\theta\}$ $\mathrm{d}\theta$ may be omitted.<br>If the integral sign and limits are not present, these can be implied by a clear attempt in later work to integrate all the terms and a substitution of limits. Students must use $2\theta$ and $5\theta$ in their working. |
| | M1 | For a minimum of 2 out of the 3 terms shown above correctly integrated.<br>Students must use $2\theta$ and $5\theta$ in their working. |
| | M1 | For substituting in the given limits the correct way around into their changed function with at least two terms, there must be at least one explicit substitution of each limit.<br>If the candidate has a constant term as part of the function, they can omit this from this step (as the terms cancel out anyway).<br>This mark may be implied by a correct answer.<br>If the candidate is working with an incorrect function and substitution of 0 clearly gives 0, this substitution does not need to be shown. |
| | A1 | For $\frac{9}{5}+\pi$ The approximate decimal value 4.941…will **usually** indicate correct working (as a guide), but a substitution of limits will be needed to get the 3rd method mark. |
| In this question, students are directed to use calculus, so must show a correct integration. The penultimate method mark may be implied from a fully correct answer, but if the answer is not correct, we must see substitution of limits as stated. | | |

## Type MS · Solving trig equations with a multiple or shifted angle

Answer 12 of 13 · Question 10 part c · 6 marks

2023 NOV · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|
| | **As with many log questions, there are a number of ways of reaching a correct answer.**<br>**The first scheme shows 3 common routes, with a set of general principles in the notes.**<br>**Not all variations in methods will have been captured, please mark to general principles, referring ANY work where this is not possible and worthy of marks to REVIEW.**<br>**This includes work that converts everything to completely new bases.** | |

Type MS · Solving trig equations with a multiple or shifted angle

Answer 13 of 13 · Question 7 part c · 2 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 7(i) (a) | $\tan 2\theta = \dfrac{\tan\theta + \tan\theta}{1 - \tan\theta\tan\theta}$ | M1 |
| | $= \dfrac{2\tan\theta}{1-\tan^2\theta}$ * | A1 cso<br>(2) |
| (b) | $1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ | |
| | $\tan^2\alpha + 2\tan\alpha - 1 = 0$ | M1 |
| | $\tan\alpha = -1 \pm \sqrt{2}$ | M1 A1<br>(3) |
| (ii) (a) | $\cos(x-30) = \sin(x+30)$ | |
| | $\Rightarrow \cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$ | M1 |
| | $\left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\cos x = \left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\sin x$ | M1 |
| | $\dfrac{\sin x}{\cos x} = \tan x = \dfrac{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}$ | M1 |
| | $\tan x = 1$ * | A1 cso<br>(4) |
| (b) | $\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2}$ | |
| | $y = -67.5^\circ, 22.5^\circ$ | M1 A1<br>(2) |
| | | **Total 11 marks** |

| Part | Mark | Notes |
|---|---|---|
| (i)(a) | **M1** | For use of $\tan(A+B) \Rightarrow \tan 2\theta = \dfrac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta}$ and simplifying to obtain the given result. |
| | **A1** | For the given result with no errors seen |
| (i)(b) | **M1** | For setting $1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ and for rearranging to obtain a 3TQ with terms in any order |
| | **M1** | For a correct attempt using a valid method to solve the 3TQ with two solutions for $\tan\alpha$ Accept use of $\theta$ for example in place of $\alpha$ |
| | **A1** | For $\tan\alpha = -1 \pm \sqrt{2}$ |
| (ii)(a) | **General principle**. (The second and third M marks can be carried out in either order)<br>First M1 – For expanding the two addition formulae **correctly and** equating.<br>Second M1 – For collecting up like terms to obtain $\sin x(.....) = \cos x(.....)$ allow sign errors.<br>Third M1 – For use of the $\tan = \dfrac{\sin}{\cos}$ identity<br>Final A1 – For the given answer with no errors | |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 13 of 13 · Question 7 part c · 2 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

| | | |
|---|---|---|
| | They can work in terms of sin30°, cos30° or tan30° or alternatively $\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{3}}$ respectively | |
| | **M1** | For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain<br>$\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$ |
| | **M1** | For collecting up like terms to obtain:<br>$\sin x(\sin 30^\circ - \cos 30^\circ) = \cos x(\sin 30^\circ - \cos 30^\circ)$<br>**NB**: This is an A mark in Epen |
| | **M1** | For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working. |
| | **A1** | For $\tan x = 1$ |
| | **ALT** | |
| | **M1** | For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain<br>$\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$ |
| | **M1** | For dividing through by $\cos x$ and collecting up like terms to obtain:<br>$\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x}$<br>$\Rightarrow \frac{\sin x}{\cos x}(\sin 30^\circ - \cos 30^\circ) = \sin 30^\circ - \cos 30^\circ$<br>**OR**<br>For dividing through by $\cos x$ **and** $\cos 30^\circ$ and also collecting up like terms to obtain:<br>$\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x\cos 30^\circ}$<br>$\Rightarrow 1 + \frac{\sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x}{\cos x} + \frac{\sin 30^\circ}{\cos 30^\circ} \Rightarrow \frac{\sin x}{\cos x}\left(\frac{\sin 30^\circ}{\cos 30^\circ} - 1\right) = \frac{\sin 30^\circ}{\cos 30^\circ} - 1$ |
| | **M1** | For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working. |
| | **A1** | For $\tan x = 1$ |
| (ii)(b) | **M1** | For solving the equation $\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2} \Rightarrow y = ..., ...$<br>(leading to two values for $y$)<br>**NB**: This is an B mark in Epen |
| | **A1** | $y = -67.5^\circ, 22.5^\circ$<br>Both values required.<br>**NB**: This is an B mark in Epen |
| | **ALT** | |
| | **M1** | For solving the equation $\tan 2y = 1$ to achieve at least one correct value for $2y$<br>$2y = \tan^{-1}1 = 45^\circ, -135^\circ$ [Accept even for example $225^\circ$ for this mark] |
| | **A1** | $y = -67.5^\circ, 22.5^\circ$<br>Both values required.<br>For extra angles in range withhold this mark.<br>Extra angles outside of the range – ignore.<br><br>**NB**: This is an B mark in Epen |

## Type MS · Solving trig equations with a multiple or shifted angle

### Answer 13 of 13 · Question 7 part c · 2 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| | | |